

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Data Mining
Program:	BSE, BCS
Submission Date:	31-03-26
Section:	6A, 8A
Assignment no. :	02
Total Marks:	60

Important Instructions:

- Use A4 sheets or narrow line sheets for solving assignments.
- Assignments done on register pages will be cancelled.
- Must attach the assignment title page shared to you at the end of assignment.
- This assignment must be completed independently by the student. The use of AI tools (including ChatGPT, Bard, Copilot, or any other generative AI system) is strictly prohibited.
- Students are expected to show all manual calculations and intermediate steps clearly.

ID3 Construction:

Important Instruction: The objective of this assignment is to understand how ID3 are constructed using different criteria including Entropy, Information Gain and Gini Index. Students will create their own dataset manually according to the given instructions.

Part 1: Dataset contains:

1. 15 to 20 rows
2. Attribute types
 - ☐ Categorical $\Rightarrow$ at least 2 attributes
 - ☐ Nominal $\Rightarrow$ at least 2 attributes
 - ☐ Binary $\Rightarrow$ at least 2 attributes
 - ☐ Target/ Class Label $\Rightarrow$ 1 attribute

Part 2: ID3 Construction Using Entropy & Information Gain

Students must:

1. Compute **Entropy of dataset**
2. Compute **Information Gain for each attribute**
3. Select attribute with **highest gain as root**
4. Split dataset
5. Repeat process recursively
6. Draw the **Tree**

Part 3: ID3 Using Gini Index

Steps

1. Compute **Gini index of the dataset**.
2. For each attribute calculate **Gini split**.
3. Select the attribute with the **lowest Gini value**.
4. Construct the **tree**.

Part 5: Comparison

Students must compare the trees generated using:

Method	Root Attribute	Tree Depth	Observations
Information Gain			
Gini Index			

Submission Requirements:

- Dataset table (same as according to the given sample) /10
- Entropy calculations /10
- Information Gain calculations /10
- Gini Index calculations /10
- Decision Tree diagrams /15
- Comparision /05

Sample dataset is given for guide(must submit according to the given two tables)

Attribute	Type	Possible Values
StudyTime	Categorical	Low, Medium, High
Attendance	Categorical	Poor, Average, Good
Major	Nominal	CS, SE, IT
Residence	Nominal	Hostel, DayScholar
Result	Class Label	Pass, Fail
InternetAccess	Binary	Yes, No

Final Dataset:

ID	StudyTime	Attendance	Major	Residence	InternetAccess	Result
1	High	Good	CS	Hostel	Yes	Pass
2	Medium	Good	SE	DayScholar	Yes	Pass
3	Low	Poor	IT	Hostel	No	Fail
4	Medium	Average	CS	DayScholar	Yes	Pass
5	Low	Poor	SE	Hostel	No	Fail

National University of Computer and Emerging Sciences

Department of Computer Science, School of Computing

DATA MINING



Assignment # _____

Section: _____

Submitted by:

Name: _____

Roll no: _____

Submitted to:

Dr. Muhammad Aasim Qureshi

Date Assigned: _____

Date Submission: _____